RISHI THAKUR

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a Computer Science foundation (M.Tech CSE, GATE CS qualified) and expertise building deep learning, NLP, and predictive systems. Experienced developing generative diffusion models (DDPM/DDIM), recurrent sequence encoders (GRU, Bi-LSTM), and transformer classifiers (BERT + XGBoost). Specialized in handling class imbalance (SMOTE + Tomek Links), custom loss formulation, and low-latency inference.

SKILLS

- **Deep Learning & Architectures:** Diffusion Models (DDPM, DDIM), Recurrent Networks (GRU, Bi-LSTM), Transformers, Multi-Head Attention, BERT Embeddings, 1D-CNN, Random Forest, XGBoost
- **ML Frameworks & Techniques:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, Resampling (SMOTE + Tomek Links), Custom Loss Formulation, Model Evaluation (F1, Precision, Recall)
- **Data Engineering & Systems:** Advanced SQL (CTEs, Window Functions, Joins), MySQL, MariaDB, Relational Schema Modeling, Python (NumPy, Pandas), Linux/Bash, Git/GitHub, Docker Basics
- **Communication & Spoken Skills:** Model Explainability to Stakeholders, Technical Writing, Research Presentations, Engineering Collaboration, Agile Communication

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Machine Learning & Data Analyst Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Architected an automated **SQL/ETL pipeline** integrating biometric punch logs (AEBAS) with State Treasury payroll records across **33 districts**.
- Engineered a **3-tier risk model** classifying compliance behaviors among bonded personnel, resolving identity discrepancies across databases.
- Conducted data preprocessing, outlier removal, and categorical normalization across disparate database architectures (MariaDB, OracleDB, e-Hospital APIs).
- Partnered with National Informatics Centre (NIC) engineers to construct unified feature tables, maintaining **100% data integrity**.
- Deployed an interactive **Power BI dashboard** with **4-tier drill-down hierarchies** to visualize predictions and utilization metrics.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Deep Generative Modeling & Telemetry*
Technologies: Python, PyTorch, GRU, DDPM/DDIM, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Architected a predictive pipeline pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** simulating 50 paths to forecast signal drops from **spatio-temporal** telemetry.
- Resolved severe **98%/2% class imbalance** via **SMOTE + Tomek Links**, training a Random Forest policy with custom loss to attain a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Applied NLP & Deep Learning*
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, TensorFlow, Scikit-learn

- Constructed dual NLP classification pipelines extracting 768-dim dense embeddings from pre-trained **BERT** with **XGBoost**, benchmarked against a deep Bi-LSTM/Transformer/1D-CNN network.
- Applied stratified sampling (80/20), text normalization (NFKD), dropout (0.5), and dynamic learning rate decay, achieving **99% Precision** and **98% Recall** in synthetic LLM text detection.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **SQL & Database Design** (Udemy)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI** (Udemy)
- **IELTS Academic:** 7.5 Band Score